

PVSPE: A pyramid vision multitask transformer network for spacecraft pose estimation

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Abstract

Spacecraft pose estimation (SPE) plays a vital role in the relative navigation system for on-orbit servicing and active debris removal. Current deep learning-based methods have made great achievements on object pose estimation. However, towards the challenging onboard SPE missions, most existing Convolutional Neural Network (CNN) methods failed to capture remote vision attention, leading to the reduction of accuracy and robustness. In this paper, we presented an end-to-end multi-task Pyramid Transformer SPE network (PVSPE) consisting of two novel feature extraction modules: EnhancedPVT (EnPVT) and SlimGFPN. The EnPVT module is designed to combine global spatial and channel attention, while the Slim GFPN module can fuse features more effectively. Matrix Fisher and multivariate Gaussian distributions are further employed to model the uncertainty of pose regression to increase its accuracy. Extensive experiments are carried out on challenging SPEED + and SHIRT datasets, to validate the performances on pose estimation and vision-based navigation, respectively. The results show that the proposed PVSPE model achieved high accuracy for SPE on the SPEED + dataset even under different scales and severe illumination, demonstrating its robustness and high generalization. Leveraging the insightful uncertainty model of PVSPE, the vision-based navigation pipeline, combined with Kalman filters, accurately estimated the satellite pose under challenging rendezvous scenarios on the SHIRT dataset, with degree-level attitude errors and centimeter-level translation accuracy at steady-state.

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1. Introduction

Spacecraft pose estimation (SPE) is essential in many space missions, such as on-orbit dock, debris detection, and other on-orbit services (OOS). In close-range ren-

dezvous and docking operations, SPE can provide the relative position and attitude of the target to the serving spacecraft (S/C) for the autonomous relative navigation system (Pirat et al., 2018; Gong et al., 2023). With that, the orbital elements of the target can be inferred, facilitating the design of collision avoidance strategies (Uriot et al., 2022). In active debris removal (ADR) missions like Clear-Space 1 (Biesbroek et al., 2021), the relative position and

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attitude are essential inputs for robotic arm control systems (Wijayatunga et al., 2023). In the past few years, SPE remained an interesting yet challenging topic due to the sharply increasing risk from space debris and obsolete spacecraft. Space-based observation and pose estimation using spaceborne sensors are therefore ever-rising in current space missions.

Spaceborne sensors typically include active sensors (LIDAR, RADAR) and passive sensors (monocular cameras, stereo cameras). Compared to active sensors, monocular cameras have relatively lower size, mass and smaller power requirement. It does not have the baseline limitation of stereo cameras, providing a wider observation range. In MEV-1 (Redd, 2020) mission, a monocular camera has been deployed to assist active sensors, to improve reliability through heterogeneous information from multiple sources. Moreover, in CubeSat formation missions (Spencer et al., 2016; Bowen et al., 2015), to comply with strict onboard constraints, monocular vision is also a promising paradigm (Cassinis et al., 2019a).

Monocular-based pose measurement commonly has two phases: initialization estimation and tracking. The initialization estimation step involves a global search without any reference from previous image frames, while the tracking phase performs a local search by combining information from current and previous frames. Current model-based tracking methods (Kelsey et al., 2006; Comellini et al., 2021) have achieved inspiring performance. However, they still depend on the robust and accurate initialization process. For initialization, traditional monocular SPE solutions commonly designed and extracted the hand-crafted features of the spatial targets, followed by feature matching to estimate the relative pose. In the case of harsh light and noise conditions in space, such hand-crafted features cannot capture the inherent features, leading to a reduction in accuracy and robustness. Due to the impressive representation and generalization capabilities of deep neural networks, an increasing number of researchers are applying deep learning (DL) methods to estimate spacecraft pose, achieving better results than those of classical handcraft models. Currently, DL-based SPE methods can be categorized into indirect multi-stage and direct end-to-end methods. Multi-stage methods need to obtain the coordinates of 2D keypoints by the prediction networks (Park et al., 2019; Cassinis et al., 2019b; Huo et al., 2020; Lotti et al., 2022). Since semantic information can improve the accuracy of location of 2D keypoints, there are some attempts to produce 2D keypoints guided with semantic segmentation (Gerard, 2019; Wang et al., 2023). After finding the keypoints, the Perspective-n-Point (PnP) solver (Lepetit et al., 2009) is applied to solve the pose by the 2D-3D correspondence. Due to the non-differentiability of PnP, it cannot be incorporated into the network structure, limiting the learning effect of the indirect method. Therefore, a promising alternative approach is to directly learn the transformation relationship between the input image and the output pose by establishing an end-to-end

pipeline (Proença and Gao, 2020; Phisannupawong et al., 2020; Posso et al., 2022; Chu et al., 2023).

Most previous methods utilized the Convolutional Neural Network (CNN) architecture as the visual backbone for extracting features. However, recent studies (Zhou et al., 2022; Mao et al., 2022; Xiong et al., 2023; Liu et al., 2023b) augured that features extracted from the pure CNN backbone may not be sufficient to resist noise and occlusion in various environments and further demonstrated the efficiency of the Vision Transformer (ViT) model due to its global exploit capabilities.

To this end, in this paper, we proposed a monocular multitask vision transformer network called PVSPE for SPE. Under the backbone-neck-head structure, a pyramid vision transformer (PVT) (Wang et al., 2021) is further optimized as the backbone (EnhancedPVT) and a novel neck (SlimGFPN) is designed for the multitask network. Specifically, we refined the feed-forward network (FFN) in PVT and added an efficient channel self-attention (ECA) (Zhou et al., 2022) module to extract more representative features. Cross-scale and cross-layer strategies are used to fuse features in the neck. In order to learn the bounding box, classification, 3D rotation and translation tasks simultaneously, we employed an Efficient Decoupled Task head, followed by a keypoint heatmap head to assist with the learning. To model the uncertainty and further improve the accuracy, probabilistic pose optimization is designed based on Matrix Fisher and multivariate Gaussian distributions. Furthermore, the PVSPE network is connected with Kalman filters to establish a vision-based navigation pipeline. To validate the efficiency of the proposed PVSPE, we conducted extensive experiments on the challenging SPEED + and SHIRT datasets. The results demonstrate that our method has superior accuracy and strong generalization.

The main contributions of this work are as follows:

1. We proposed a novel multi-task vision transformer SPE framework by designing two feature extraction and fusion modules: Enhanced PVT and SlimGFPN.
2. We drew from Matrix Fisher and Multivariate Gaussian distributions to improve the accuracy of pose regression and construct vision-based navigation.
3. The experimental results on the challenging SPEED + and SHIRT datasets show the effectiveness and robustness of the proposed PVSPE.

2. Related work

2.1. Monocular vision spacecraft pose estimation

The most intuitive solution for monocular-based SPE is to employ convolution neural networks (CNNs) for target detection and 2D keypoints prediction, and then utilize the PnP algorithm (Gao et al., 2003) for pose estimation. Some pioneers have conducted such attempts based on many

well-known models. Park et al. (2019) combined the darknet (Redmon and Farhadi, 2018) and the mobilenet (Howard et al., 2017) for the detection of spacecraft and the regression of key points. Huo et al. (2020) designed a keypoint existence judgment model embedded in YOLOv3 network to obtain the keypoints. To accelerate SPE inference, Lotti et al. (2022) adopted the efficient-lite network (Tan and Le, 2019) for keypoint regression. Gerard (2019) and Wang et al. (2023) further applied semantic segmentation to improve the accuracy of keypoint location. Under such a CNN-PnP framework, those indirect method seems to be the preferred solution, since the current state-of-the-art detection network is far more the GOAT. However, the non-differentiability of the PnP algorithm directly limits the learning effect of the multistage methods.

Compared with above indirect methods, the end-to-end solutions directly predicted 3D rotation and translation using deep neural networks rather than the mid-level-feature representation such as semantic keypoints. Unlike 3D translation, the learning of 3D rotations is inherently more challenging due to differences in topology between $\mathbb{R}^N$ and 3D special orthogonal group ($SO(3)$) (Zhou et al., 2019). Previous studies like SPN (Sharma et al., 2018) and UrsoNet (Proença and Gao, 2020) tried to discretize the 3D rotation into bins. However, such discretization inevitably brought discontinuous supervision information to the neural network, leading to failure of rotation learning (Zhou et al., 2019). To ensure rotation continuity, Park (Park and D'Amico, 2022) and Chu (Chu et al., 2023) adopted the 6D (Zhou et al., 2019) representation and the 9D representation (Levinson et al., 2020), respectively. Inspired by such high-dimensional approaches, in this paper, we followed this type of solution but utilized probabilistic model based on the Matrix Fisher distribution to improve the accuracy of 3D rotation regression.

2.2. Vision transformer (ViT) backbone

Benefiting by the excellent global information interaction ability of the self-attention mechanism, the ViT backbone has achieved impressive performance in various vision tasks, such as semantic segmentation (Xie et al., 2021; Xiong et al., 2023) and object detection (Liu et al., 2021; Wang et al., 2021). As the complexity of vanilla self-attention in ViT has reached $\mathcal{O}(n^2)$, many studies have tried to design more efficient self-attention mechanisms like Swin (Liu et al., 2021) and PVT (Wang et al., 2021). Swin restricts the span of the self-attention to local spatial windows. PVT created spatial reduction attention (SRA) by cutting the size of K and V in the self-attention block. In addition to these spatial dimension cutting methods, Zhou et al. (2022) demonstrated the importance and efficiency of global channel information interaction and proposed an efficient channel self-attention (ECA) mechanism.

2.3. Vision feature pyramid

For vision-based multitask frameworks, the neck is designed to aggregate different scale features extracted from the backbone. The pyramid structure of the neck has been confirmed to be more effective in feature fusion in object detection missions (Xu et al., 2022a) and semantic segmentation missions (Xiong et al., 2023; Liu et al., 2023a). Liu (Liu et al., 2018) designed the PAFPN model that introduced a bi-directional information structure consisting of both top-down and bottom-up pathways to fuse multi-scale features. For efficiency, Tan et al. (2020) improved this bi-directional structure by deleting nodes with only one input edge in their proposed BiFPN model. In BiFPN, multiple layers were stacked together by skip connection. Such a "Stack" mode would increase the multitask-fitting ability of the network, as well as the generalization performance. Recently, Tan et al. (2021) designed a new "Stack" FPN, called GFPN, by merging features of the upper and current layers to promote information exchange. Followed by GFPN, Xu et al. (2022b) further optimized cross-layer feature transmission and fusion to perform more accurate object detection.

3. Monocular relative navigation pipeline

The proposed vision-based spacecraft relative navigation pipeline is presented in Fig. 1. It is composed of two modules, the proposed PVSPE module and the navigation filter module. The PVSPE network extracted features from the image sequence to estimate the relative translation T and the attitude parameters R of the target in the camera frame. Then the estimation results are sent to the navigation filter. Since the estimation of the relative translation is commonly more robust with respect to the estimation of relative attitude (Opromolla et al., 2017), to improve stability, we adopted a decoupling strategy of navigation filters.

4. Pose estimation network

In this section, we will first introduce the proposed PVSPE network and illustrate the proposed Enhanced PVT (EnPVT) backbone and Slim GFPN module, as well as the efficient task head and multitask loss for robust and accurate pose estimation of spacecraft.

4.1. Overall framework

The overall framework of the proposed PVSPE model is shown in Fig. 2. Given a spacecraft image, PVSPE can predict its class, bounding box, 3D rotation and translation, simultaneously.

Given a spacecraft image, the EnPVT backbone will perform a 6-stage feature extraction and totally generate

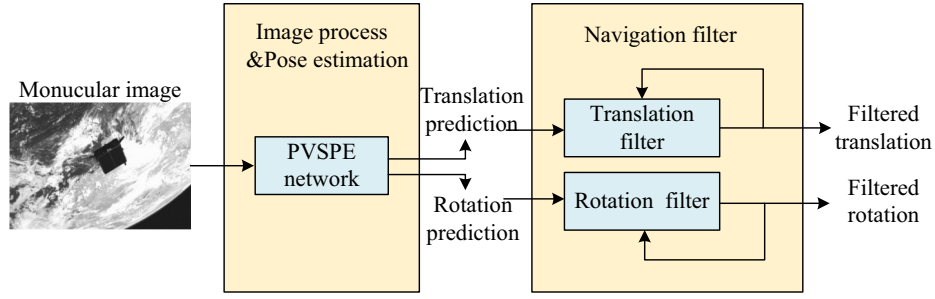


Fig. 1. Illustration of the proposed vision-based pipeline for spacecraft monocular relative navigation.

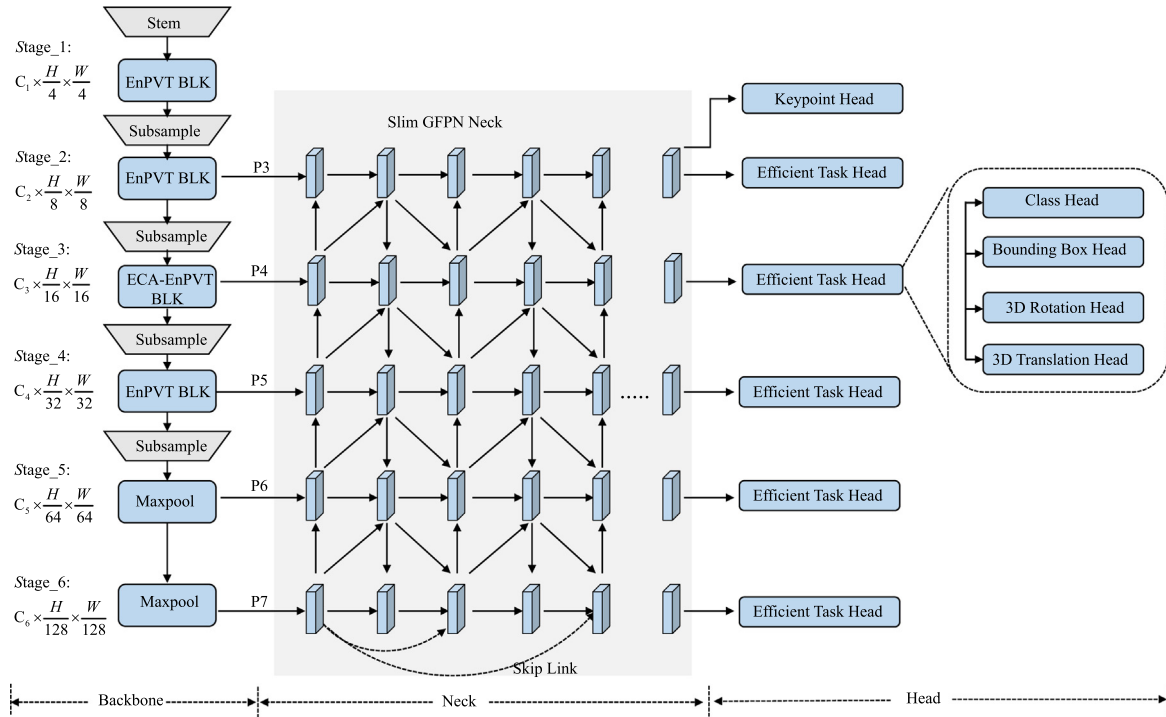


Fig. 2. The proposed SPE framework consists of four main modules: Enhanced PVT backbone, Slim GFPN, Efficient task head, and keypoint heatmap auxiliary head.

$\{P_i \in \mathbb{R}^{C_i \times H_i \times W_i}\}_{i=2}^7$ level features. For the first four stages, an overlap patch embedding (Opm) (Xie et al., 2021) block is combined with EnPVT, where Opm downsampled the features of the previous stages and the EnPVT block was repeated to further enhance the feature extraction capability. In stage 3, the ECA (Zhou et al., 2022) module is embedded into EnPVT for the channel information interaction. Two Maxpool operations are carried out at the end of the backbone, producing the P_6 and P_7 level features.

For the neck, the proposed Slim GFPN neck will further fuse and refine the features of the backbone. Both cross-scale and cross-layer fusion strategies are used for efficient information exchange and integration, and skip-link is applied to enhance the gradient propagation. The hierarchical features generated by Slim GFPN are finally transferred to the efficient task head to conduct multitask

prediction. Only P_3 level features, for its higher-resolution requirement, are used for the auxiliary head of the keypoint heatmap.

4.2. EnPVT backbone

As shown in Fig. 3 (a), the original PVT structure consists of SRA and FFN blocks. Only the Linear layer and GELU (Hendrycks and Gimpel, 2016) activation function are used in FFN (Fig. 5 (a)), resulting in the failure of local feature interaction. Therefore, inspired by LocalVit (Li et al., 2021) and Segformer (Xie et al., 2021), we design a ReFFN by introducing a 3×3 depth-wise convolution with skip connection to enhance such locality, as shown in Fig. 5 (b). In addition, SRA only established global spatial attention on all input tokens, the transformer block thus cannot capture the remote interaction of features

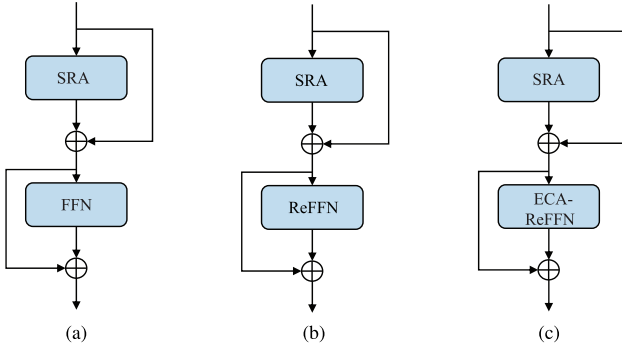


Fig. 3. The structure of transformer block. (a) Original PVT Block (b) The proposed EnPVT block (c) The proposed ECA-EnPVT block.

along the channel dimension. For more efficient feature interaction, we embedded the ECA (Zhou et al., 2022) block into EnPVT to perform long-range channel dependency, producing the final ECA-EnPVT structure, as shown in Fig. 3 (c).

4.2.1. SRA: Spatial reduction self-attention

The structure of SRA is shown in Fig. 4 (a). The input feature is separated into three vectors Q , K and $V \in R^{N \times C}$. Reduction convolution $Conv_{R \times R}$ is applied to reduce the size of K and Q , generating K_n and Q_n with $R^{N_n \times C}$. The spatial self-attention can be formulated as:

$$[Q_n, K_n] = LN(Conv_{R \times R}([Q, K])), \quad (1)$$

$$Attention(Q_n, K_n, V) = Softmax\left(\frac{Q_n K_n^T}{\sqrt{d_{head}}}\right)V, \quad (2)$$

where LN is the layer normalization. After reduction convolution, SRA performs the dot-product operation of Q_n and K_n^T over the spatial dimension ($N_n \times N_n$). The computational complexity is thus reduced to $O\left(\frac{N^2}{R^2}\right)$, leading to the efficiency of SRA in higher-resolution applications.

4.2.2. ECA: Efficient channel self-attention

The structure of ECA is shown in Fig. 4 (b). The Softmax function is utilized to normalize Q and K . Moreover,

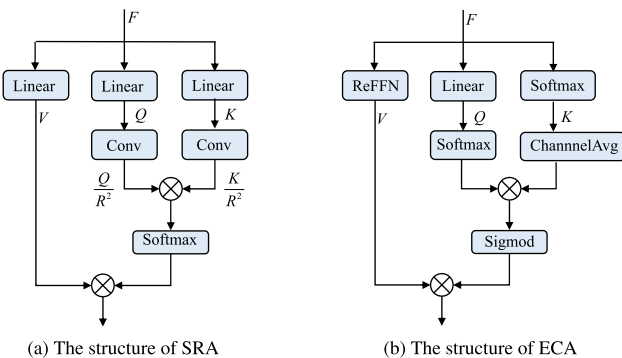


Fig. 4. The structures of SRA and ECA.

for K , Adaptive Averaging Pooling (AVG) is applied to aggregate channel information to generate prototypes for each channel, producing K_n . The ECA can be formulated as

$$K_n = AVG(K), \quad (3)$$

$$Attention(Q, K_n, V) = Sigmoid\left(\frac{Q^T K_n}{\sqrt{d_{head}}}\right)V, \quad (4)$$

ECA computes the dot product of Q^T and K_n along the channel dimension ($C \times C$), yielding the attention weights matrix. Then, the Sigmoid function is used to normalize the attention weights matrix.

4.2.3. Refined FFN

To capture more of the local information, we proposed a Refined FFN (ReFFN) structure as shown in Fig. 5 (b). In ReFFN, 3×3 depth-wise convolution is utilized to create the local context in EnPVT, since it has a small feature reception range and is more sensitive to local information such as textures and boundaries (Ding et al., 2022). In addition, the combination of skip connection and LN can stabilize the training of neural networks to improve accuracy (Liu et al., 2020), especially for Transformer Architectures (Vaswani et al., 2017). Therefore, the skip connection is added before the LN layer. The ReFFN can be formulated as:

$$\begin{aligned} X_1 &= Linear(X_{in}) \\ X_2 &= X_1 + DWConv_{3 \times 3}(X_1) \\ X_3 &= GELU(LN(X_2)) \\ X_{out} &= Linear(X_3), \end{aligned} \quad (5)$$

4.3. Slim GFPN neck

Considering onboard requirements, we optimized the original GFPN to develop SlimGFPN (SGFPN) for more effective feature fusion. The structure of SlimGFPN (SGFPN) and SGFPN fusion block $\mathcal{F}$ (SFB) is shown in Fig. 6 (a) and (b).

SGFPN takes five-level feature maps (ie, $P_i (i = 3, 4, 5, 6, 7)$) from the backbone to handle large-scale variations of targets in space. Motivated by EGFPN (Xu et al., 2022b) on object detection, we explore the impacts of cross-layer up-samplings (CSUs) and down-samplings (CSDs) on pose estimation for spacecraft by ablation studies. As shown in Table 2, we find that the CSU even brought a negative contribution to the pose estimation, indicating that excessive connections may cause overfitting or optimization issues.

Based on that, SGFPN only adopted CSDs. After feature transmission, the effective SFB is employed to merge features. As depicted in Fig. 6 (b), the input features are combined and then subjected to a 1×1 convolution operation to decrease the channel, followed by a 3×3 convolution to extract both semantic and spatial information. To

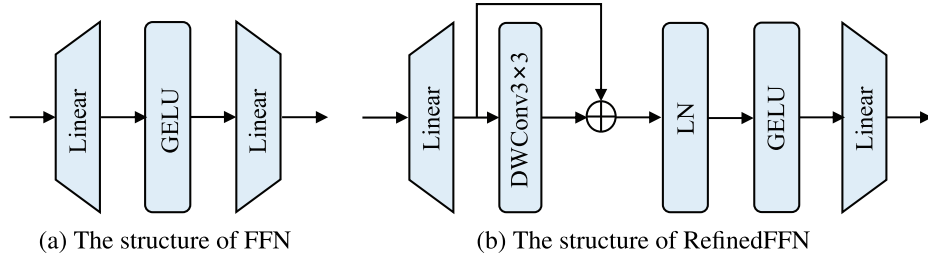
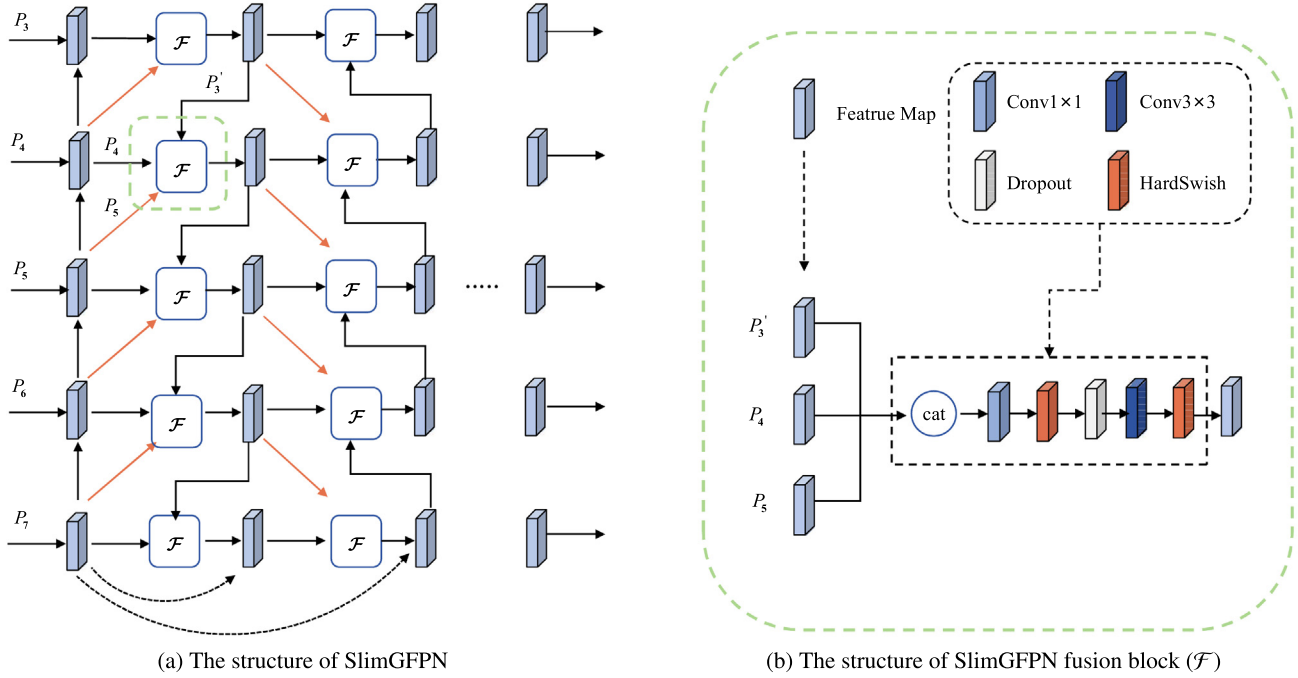


Fig. 5. FFN vs Proposed RefinedFFN.

Fig. 6. Illustration of SlimGFPN and SlimGFPN fusion block ($\mathcal{F}$).

address potential noise and occlusion in the space, dropout is employed to reinforce the model's robustness and generalizability. Furthermore, activation functions are implemented using HardSwish (Howard et al., 2019) to enhance nonlinearity and efficiency. Take the fusion of feature $P_{3'}$, P_4 and P_5 as an example, the SFB can be expressed as follows:

$$\begin{aligned}
 F_1 &= \text{Concat}(P_i(i = 3', 4, 5)) \\
 F_2 &= \text{HardSwish}(\text{Conv}_{1 \times 1}(F_1)) \\
 F_3 &= \text{Dropout}(F_2) \\
 F_{out} &= \text{HardSwish}(\text{Conv}_{3 \times 3}(F_3))
 \end{aligned} \tag{6}$$

4.4. Efficient task head

For multitask learning, the conflict among different tasks remains an interesting yet challenging topic, which always leads to non-convergence or accuracy issues. One type of solution is to utilize a decoupled head, which has been substantiated by TOOD (Feng et al., 2021) and PPYOLOE (Xu et al., 2022a). Therefore, we employed

the efficient task decoupled head as shown in Fig. 7. In Efficient Task decoupled Head, the efficient squeeze-and-excitation (ESE) (Lee and Park, 2020) block (Fig. 7 (a)) is embedded in every task head. The ESE module generates channel-specific descriptors by compressing the feature maps per channel through global average pooling (GAP). ESE is defined as:

$$W_C = \text{Sigmoid}(\text{FC}(\text{GAP}(X))), \tag{7}$$

$$X_{refine} = \text{Conv}_{1 \times 1}(XW_C), \tag{8}$$

where $X \in \mathbb{R}^{C \times W \times H}$ is the feature from the Slim GFPN neck, GAP is the global average pooling, W_C is the channel weight, X_{refine} is the refined feature. In addition, since object class, pose information are more difficult to optimize than object bounding boxes, a shortcut is used in corresponding task branches.

4.5. Multi task loss function

For the multitask framework beyond detection, the loss function should be composed of several aspects, including bounding box regression loss L_{bbox} , classification loss L_{cls} ,

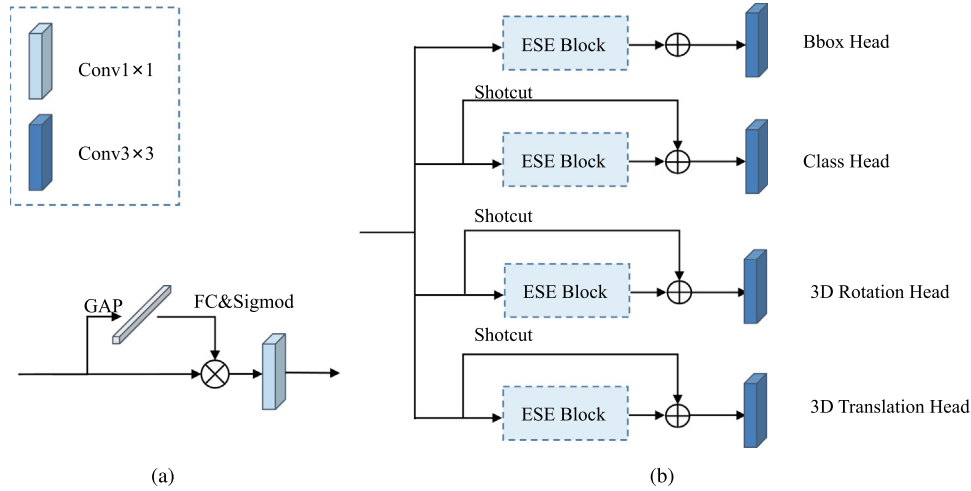


Fig. 7. (a) Efficient Squeeze-and-Excitation (ESE) Block. (b) Efficient task head.

3D rotation loss L_{rot} and translation loss L_{tra} , as well as the key-point loss L_{kpt} . Therefore, the total loss can be directly formulated as the sum of all the above losses:

$$L_{total} = w_1 L_{cls} + w_2 L_{bbox} + w_3 L_{rot} + w_4 L_{tra} + w_5 L_{kpt}, \quad (9)$$

where $w_i (i = 1, 2, \dots, 5)$ are the tuning parameters for each loss. In order to supervise spacecraft classification, we utilize Focal Loss (FL), which can improve the accuracy of classification especially for hard-to-classify samples (Lin et al., 2017). For spacecraft detection, the loss of complete intersection-over-union (CIoU) (Zheng et al., 2020) is adopted to consider the aspect ratio of the spacecraft bounding box. For the keypoint heatmap, the loss of Gaussian format is preferred due to its efficiency in key-point prediction.

In space, noise and light reflection may cause ambiguity in rotation estimation of the spacecraft. In recent work (Mohlin et al., 2020), Matrix Fisher (Khatri and Mardia, 1977) distribution $\mathcal{MF}(\mathbf{R}; \mathbf{F})$ has been demonstrated to be effective in modeling uncertainty to reduce the adverse effects of ambiguity. Therefore, we utilize the Matrix Fisher distribution to increase accuracy, generating a negative log-likelihood (NLL) type of L_{rot} :

$$L_{rot} = -\log(p(\mathbf{R}|\mathbf{F})), \quad (10)$$

$p(\mathbf{R}|\mathbf{F})$ denotes the probability density function:

$$p(\mathbf{R}|\mathbf{F}) = \mathcal{MF}(\mathbf{R}; \mathbf{F}) = \frac{1}{c(\mathbf{F})} \exp(\text{tr}(\mathbf{F}^T \mathbf{R})), \quad (11)$$

where $\mathbf{R} \in SO(3)$ is a matrix representing the 3D rotation truth, $c(\mathbf{F})$ denotes the normalizing constant with $\mathbf{F} \in \mathbb{R}^{3 \times 3}$ as the unconstrained matrix parametrizing the distribution. By SVD decomposition of predicted $\mathbf{F}$, the final 3D rotation prediction $\hat{\mathbf{R}}$ can be derived. For more details on Matrix Fisher and NLL, please refer to (Mohlin et al., 2020).

Similarly, the 3D translation can be modeled by Multivariate Gaussian distribution $\mathcal{MG}(\mathbf{T}; \Sigma)$. The NLL format of L_{tra} is thus defined as:

$$L_{tra} = -\log(p(\mathbf{T}|\Sigma)), \quad (12)$$

Likewise, $p(\mathbf{T}|\Sigma)$ denotes the probability density function:

$$p(\mathbf{T}|\Sigma) = \mathcal{MG}(\mathbf{T}; \Sigma) = \frac{\exp\left(-\frac{1}{2}(\mathbf{T} - \hat{\mathbf{T}})^T \Sigma^{-1}(\mathbf{T} - \hat{\mathbf{T}})\right)}{(2\pi)^{3/2} |\Sigma|^{1/2}}. \quad (13)$$

where $T = [x, y, z]^T$ is the 3D translation vector and $\Sigma \in \mathbb{R}^{3 \times 3}$ denotes the covariance matrix associated with T . By minimizing L_{tran} , we finally get the optimal prediction $\hat{T}$ and Σ .

5. Navigation filter

5.1. Translation filter

The Extended Kalman Filter (EKF) is adopted to improve relative translation estimation. To utilize orbital dynamics to propagate states, the translation vector T in the camera frame is transformed into the servicer's Local Vertical Local Horizontal (LVLH) frame. Thus, the state vector S_T of EKF is composed of the position vector $\rho = [x_L, y_L, z_L]^T$ and velocity $v = [\dot{x}_L, \dot{y}_L, \dot{z}_L]^T$ in LVLH, given by:

$$S_T = [x_L, y_L, z_L, \dot{x}_L, \dot{y}_L, \dot{z}_L]^T \quad (14)$$

In the LVLH frame, the dynamic model of the EKF is expressed by the following nonlinear differential equations (De Ruiter et al., 2012):

$$\ddot{x}_L = 2\dot{\theta}_s \dot{y}_L + \ddot{\theta}_s y_L + \dot{\theta}_s^2 x_L + \frac{\mu}{r_s^2} - \frac{\mu(r_s + x_L)}{[(r_s + x_L)^2 + y_L^2 + z_L^2]^{3/2}} \quad (15)$$

$$\ddot{y}_L = -2\dot{\theta}_s \dot{x}_L - \ddot{\theta}_s x_L + \dot{\theta}_s^2 y_L - \frac{\mu y_L}{[(r_s + x_L)^2 + y_L^2 + z_L^2]^{3/2}} \quad (16)$$

$$\dot{z}_L = -\frac{\mu z_L}{\left[(r_s + x_L)^2 + y_L^2 + z_L^2\right]^{3/2}} \quad (17)$$

where μ is Earth's gravitational constant, θ_s is the servicer true anomaly, and r_s is in the orbital position magnitude of the servicer in the inertial space. The EKF adapts the process noise of the dynamic model using the maximum likelihood estimation method (Fraser and Ulrich, 2021). The measurement vector Y_T of EKF is expressed by:

$$Y_T = \text{diag}[\rho, v] + [Cov_\rho, Cov_v] \quad (18)$$

The rotation matrix of the camera to the LVLH frame is defined as M_{CL} . Position vector ρ and measurement covariance Cov_ρ is transformed from PVPSE prediction:

$$\rho = M_{CL}T \quad (19)$$

$$Cov_\rho = \Sigma \quad (20)$$

5.2. Rotation filter

Since higher non-linearities of rotation problems, the Unscented Kalman Filter (UKF) (Crassidis and Markley, 2003) is applied to improve rotation prediction. The Modified Rodrigues Parameter (MRP) represents quaternion error for prediction propagation and measurement fusion. The error-state vector S_R is given by:

$$S_R = [\delta p_r, w_r]^T \quad (21)$$

$$\delta p_r = \frac{4}{1 + \delta q_{r0}} \delta q_{r1-3} \quad (22)$$

where δp_r is MRP vector. q_r and w_r are the relative quaternion and angular velocity of the target in the servicer frame. The relative rotational kinematic and dynamic model of UKF can be formulated as (Capuano et al., 2018; Pesce et al., 2019):

$$\dot{q}_r = \frac{1}{2} \Omega(\omega_r) q_r + Cov_p \quad (23)$$

$$J_s \dot{\omega}_r = J_s R J_r^{-1} [m_t - \omega_t \times J_t \omega_t - J_s \omega_s \times \omega_s - [m_s - \omega_s \times J_s \omega_s]] \quad (24)$$

where R is the relative rotation matrix of the target in the camera frame. ω_t and ω_s are target and servicer absolute angular velocity, respectively. J_t and J_s is the target and servicer inertia matrix, respectively. Note that the control moment m_t of the noncooperative target is 0. Cov_p is the process noise matrix, which is adjusted through the covariance matching method. The measurement vector Y_R of UKF is expressed by:

$$Y_R = [R, \omega_r] \oplus [Cov_R, Cov_\omega] \quad (25)$$

R and its covariance Cov_R are derived from Matrix Fisher parameter matrix $\mathbf{F}$ of PVSPE network (Markley and Mortari, 2000):

$$U \text{diag}[s_1, s_2, s_3] V^T = \mathbf{F} \quad (26)$$

$$R = U \text{diag}[1, 1, \det(UV)] V^T \quad (27)$$

$$Cov_R = U \text{diag}[(s_2 + s_3)^{-1} (s_3 + s_1)^{-1} (s_1 + s_2)^{-1}] U^T \quad (28)$$

6. Experiments

In this section, we will first introduce the dataset, evaluation criteria and training setup. Then, extensive ablation studies are carried out to validate the roles of the proposed modules. Next, we compare the performance of the proposed PVSPE with previous state-of-the-art methods on the SPEED+ dataset and quantify the prediction uncertainty, as well as its runtime on two edge computing equipment. Finally, the performance of the PVPSE-based navigation pipeline is evaluated on the SHIRT dataset.

6.1. Datasets

We conduct experiments on the SPEED+ (Park et al., 2022) dataset that contains three domains, synthetic, lightbox and sunlamp image sets. The resolution of the image is 1920×1200 . The lightbox and sunlamp domains are captured in Hardware-in-the-Loop (HIL) environment to simulate real Earth diffuse reflection and strong direct sunlight, respectively. The synthetic domain consists of 59660 images generated by OpenGL, while lightbox and sunlamp domains have 6740 and 2791 shots, respectively. In this work, we use the synthetic domain to train our proposed network. Then, the lightbox and sunlamp domains are used to evaluate our method.

6.2. Implementation details

6.2.1. Training settings

Our model is trained on an NVIDIA GeForce RTX 3090 GPU and implemented by Pytorch. The input image is resized to 768×512 . AdamW algorithm is used with a basic learning rate of 0.0001 and a batch size of 8. The model is trained for 100 epochs, and the learning rate is reduced to 0.00005 and 0.00001 in the 20th and 40th epochs, respectively. Multiple strategies of data enhancement are utilized to obtain better generalization. The weight of each loss is set to $w_i (i = 1, 2, 3, 5) = 1.0$ and $w_4 = 0.25$.

6.2.2. Evaluation criteria

For the 2D bounding box prediction, the Intersection over Union (IoU) metric is used. The 3D translation normalization error E_T and rotation error E_R are used to evaluate the accuracy of SPE:

$$\begin{aligned} E_R &= 2 \times \arccos|\langle \hat{q}, q \rangle| \\ E_T &= \frac{\|t - \hat{t}\|}{t} \\ E_{Pose} &= E_R + E_T, \end{aligned} \quad (29)$$

where E_{Pose} denotes the total pose error (Kisantani et al., 2020) of combined rotation and translation, t is the trans-

lation vector, and q stands for the quaternion of rotation. According to the calibration results of the HIL facilities, if $E_R < 0.002950(0.169^\circ)$ and $E_T < 0.002173$, they are set to zero.

6.3. Ablation studies

We conduct extensive ablation studies to verify the effectiveness of each component of our approach. PVTb1 (Wang et al., 2021) is selected as the baseline. For the first four stages, the transformer block is repeated 2 times ($L_i(i = 1, 2, 3, 4) = [2, 2, 2, 2]$). The fifth and sixth stages use simple max-pooling for downsampling. The channel dimensions for the six stages are set to $[64, 128, 320, 512, 256, 256]$. The Slim GFPN is set as the neck. The depth and width are 9 and $256 \times \phi_w$, where ϕ_w is 0.8.

6.3.1. Effect of the Enhanced PVT backbone

Based on PVTb1, we carry out further experiments to validate the impact of Refined-FFN and ECA. The original FFN is first replaced by Refined-FFN and then ECA is embedded. As shown in Table 1, the substitute of the original FFN by Refined FFN has made the E_{pose} dropping from 0.154 to 0.133 on the lightbox test domain, and on the sunlamp test domain from 0.257 to 0.228, indicating its efficiency. The joining of ECA further reduced E_{pose} from 0.133 to 0.101 on the lightbox test domain, and from 0.228 to 0.178 on sunlamp test domain. The higher accuracy demonstrated the superiority of ECA and the importance of long-range channel dependency.

6.3.2. Effect of Slim GFPN neck

For the argument about the efficiency of CSU and CSD on cross-layer information interactions, the functionalities of both of them are shown in Table 2. As the result on lightbox domain, CSUs and CSDs reduced E_{pose} from 0.148 to 0.134 and 0.101, respectively. On sunlamp, errors dropped from 0.237 to 0.226 and 0.178, respectively. However, with the combination of CSUs and CSDs, the pose accuracy dramatically decreased compared to CSDs alone. This is mainly because too many feature connections will lead to feature redundancy and further submerge the expression of valid features. Besides, CSUs brought out more parameters ($\approx 3M$) and computational burden ($\approx 6G$) of the network, which made the optimization harder. Therefore, only CSDs are used in our Slim GFPN module. Furthermore, to better evaluate the performance

of the proposed Slim GFPN, it is compared with EGFPN (Xu et al., 2022b). As shown in Table 3, the results demonstrate the superiority of SGFPN on pose estimation.

6.3.3. Effect of the keypoint auxiliary Head

The effects of the keypoint auxiliary head (kpt-aux-head) are shown in Table 4. We utilized keypoint heatmaps to help with and guide the learning of pose regression. The kpt-aux-head can reduce E_{pose} on the lightbox domain from 0.143 to 0.101, and from 0.216 to 0.178 on sunlamp domain. As shown in Fig. 8, kpt-aux-head speeds up the convergence of the model in all test domains. This demonstrated that assisted learning of semantic keypoint features can reduce learning difficulty and thus improve model accuracy. Since kpt-aux-head is only used during training, parameters and FLOPs are not measured in this paper.

6.3.4. Effect of representations for 3D rotation and translation

Based on the above network structure, for rotation, 6D (Zhou et al., 2019), 9D (Levinson et al., 2020) and Matrix-Fisher distribution (Mohlin et al., 2020) are taken into comparison. For translation, we compared direct regression and Multivariate Gaussian distribution. The impact of the above representations for 3D translation and rotation are displayed in Table 6 and Table 5. Compared with 9D rotation, Matrix Fisher-based representation of rotation brought out a lower E_{pose} , reducing from 0.122 to 0.101 on lightbox, and from 0.222 to 0.178 on sunlamp test domain, as shown in Table 5. For 3D translation, compared with the direct regression-based method, the Multivariate Gaussian-based method cut down the E_{pose} to 0.101 on lightbox, and 0.178 on sunlamp as displayed in Table 6. Such results indicate its efficiency of probability distribution-based regression.

6.4. Comparisons with state-of-the-art methods

We compared the proposed PVSPE with some existing state-of-the-art methods, including KRN (Park et al., 2019), SPNv2 (Park and D'Amico, 2022) and STGC (Wang et al., 2022). SPNv2 utilized AdaBN (Li et al., 2018) and entropy minimization for online test domain refinement to improve the accuracy. STGC brought the geometrical constraints into semantic segmentation to conduct ST (self-training) on the test domain. However, in real applications, it is difficult to conduct test domain refine-

Table 1
Effect of Refined-FFN and ECA in the proposed Enhanced PVT on the SPEED + dataset.

Backbone	Params (M)	FLOPs (G)	lightbox				sunlamp			
			IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]
PVTb1	69.93	97.74	0.916	0.020	0.134	0.154	0.914	0.033	0.224	0.257
+ RefinedFFN	69.95	97.79	0.917	0.019	0.114	0.133	0.916	0.027	0.201	0.228
+ ECA	70.15	98.10	0.919	0.017	0.084	0.101	0.917	0.022	0.156	0.178

Table 2

Effect of CSUs and CSDs on cross-layer information interactions in the neck on the SPEED + dataset.

CSDs	CSUs	Params (M)	FLOPs (G)	lightbox				sunlamp			
				IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]
		67.24	96.64	0.915	0.020	0.128	0.148	0.914	0.030	0.207	0.237
	✓	70.18	102.66	0.915	0.018	0.116	0.134	0.914	0.028	0.198	0.226
✓		70.15	98.10	0.919	0.017	0.084	0.101	0.917	0.022	0.156	0.178
✓	✓	73.09	104.12	0.916	0.017	0.095	0.112	0.915	0.026	0.183	0.209

Table 3

Comparison of performance for SlimGFPN and EGFPN on SPEED + dataset.

Method	lightbox			sunlamp		
	E_T [-]	E_R [rad]	E_{pose} [-]	E_T [-]	E_R [rad]	E_{pose} [-]
EGFPN (Xu et al., 2022b)	0.019	0.108	0.127	0.031	0.183	0.214
SlimGFPN (Proposed)	0.017	0.084	0.101	0.022	0.156	0.178

Table 4

Ablation experiment of the keypoint auxiliary head on the SPEED + dataset.

Method	lightbox				sunlamp			
	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]
w/o kpt-aux-head	0.915	0.019	0.124	0.143	0.914	0.026	0.190	0.216
w kpt-aux-head	0.919	0.017	0.084	0.101	0.917	0.022	0.156	0.178

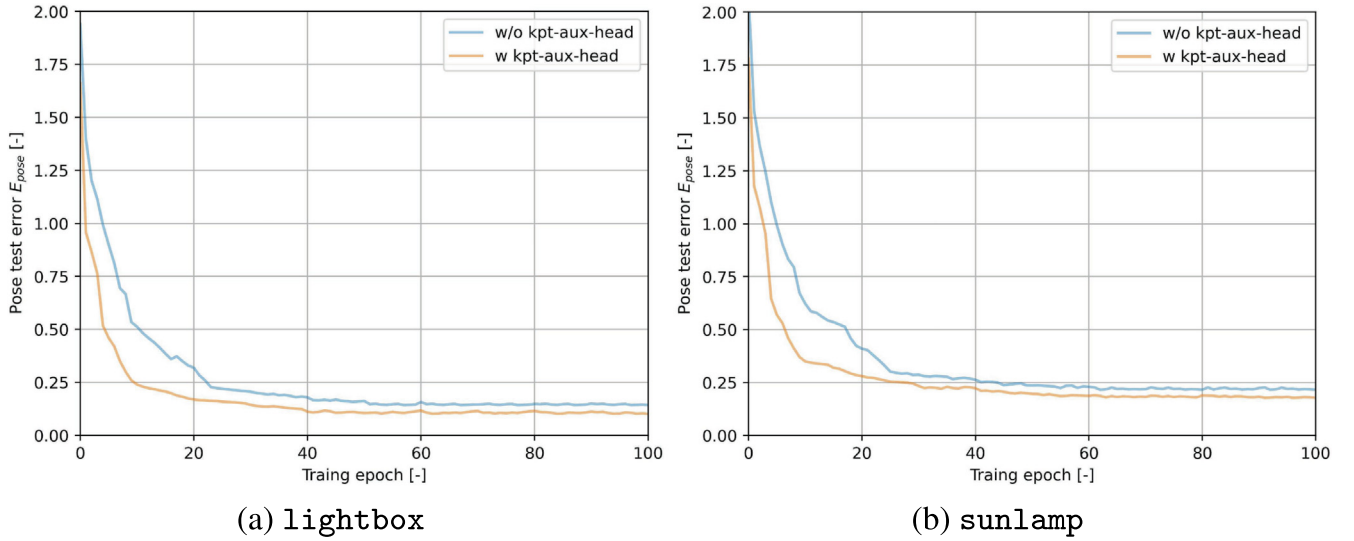
Fig. 8. Pose error E_{pose} curves of model with and without kpt-aux-head on the lightbox and sunlamp test domain. The vertical axis represents the pose test error E_{pose} . The horizontal axis represents the training epoch.

Table 5

Ablation of 3D rotation representations on the SPEED + dataset.

3D rotation	lightbox				sunlamp			
	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]
6D (Zhou et al., 2019)	0.917	0.022	0.116	0.138	0.913	0.027	0.221	0.248
9D (Levinson et al., 2020)	0.917	0.019	0.103	0.122	0.914	0.025	0.197	0.222
Matrix-Fisher (Mohlin et al., 2020)	0.919	0.017	0.084	0.101	0.917	0.022	0.156	0.178

Table 6
Ablation of 3D translation representations on the SPEED + dataset.

3D translation	lightbox				sunlamp			
	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]	IoU[-]	E_T [-]	E_R [rad]	E_{pose} [-]
Direct regression	0.918	0.023	0.092	0.115	0.915	0.030	0.169	0.199
Multivariate Gaussian	0.919	0.017	0.084	0.101	0.917	0.022	0.156	0.178

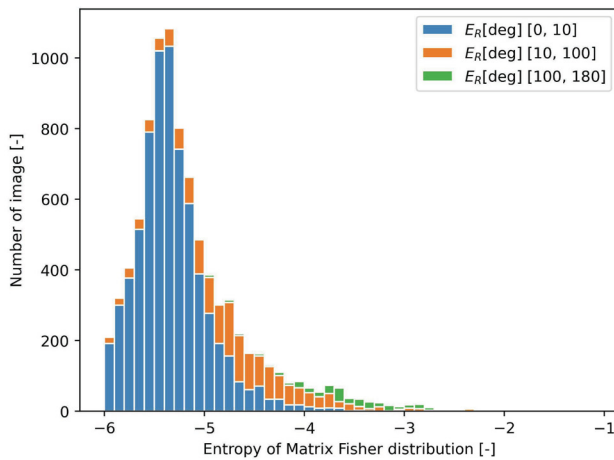
Table 7
Comparisons with state-of-the-art methods on the SPEED + dataset.

Method	lightbox			sunlamp		
	E_T [-]	E_R [rad]	E_{pose} [-]	E_T [-]	E_R [rad]	E_{pose} [-]
KRN (Park et al., 2019)	0.179	0.631	0.810	0.223	1.097	1.320
KRN (Oracle) ¹	0.043	0.107	0.150	0.037	0.093	0.130
SPNv2 (Park and D'Amico, 2022) ²	0.024	0.098	0.122	0.030	0.168	0.198
STGC (Wang et al., 2023) ³	0.046	0.116	0.162	0.006	0.047	0.053
PVSPE (Ours)	0.017	0.084	0.101	<u>0.022</u>	0.156	0.178

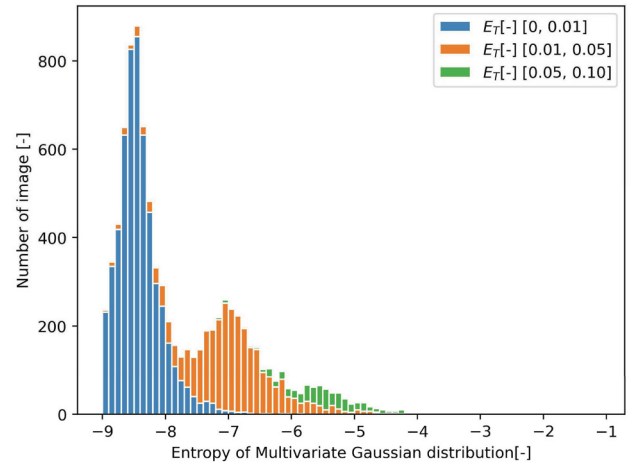
¹ It is trained directly on the test domain images and labels.

² It conducts test domain refinement on the test domain.

³ It conducts ST (self-training) on the test domain.



(a) The relationship between rotation error (E_R) and entropy (H_R)



(b) The relationship between translation error (E_T) and entropy (H_T)

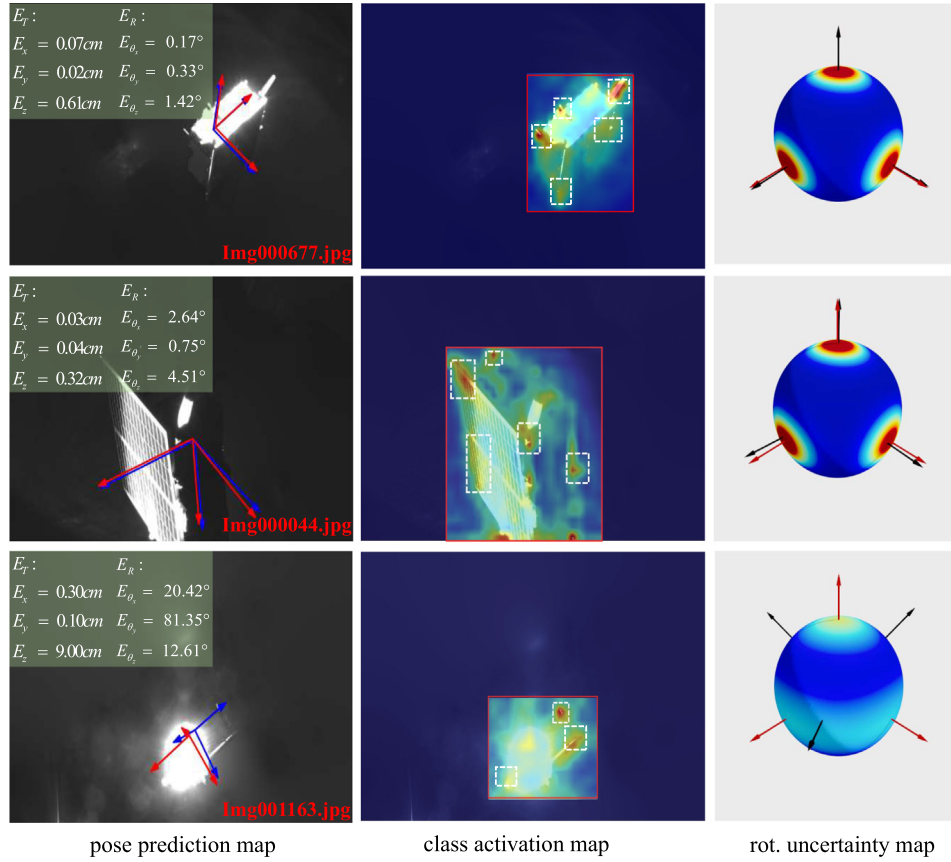
Fig. 9. Visualization of the relationship between pose error and the corresponding distribution entropy when tested on the SPEED + test set. The vertical axis represents the number of images. The horizontal axis represents the entropy of Matrix Fisher or Multivariate Gaussian distribution. The color coded by the rotation or translation errors.

ment (training) in a complicated and volatile space environment. Therefore, PVSPE focused on improving long-range modeling and multi-scale information interaction to enhance the generalization and accuracy of the model on the test domain. The comparison results of different methods are shown in Table 7. Without any test domain refinement, on the lightbox test domain, PVSPE can still outperform STGC and SPNv2, even surpass the Oracle KRN, which is trained using the test domain annotations and images directly. On more challenging sunlamp test domain, our approach still achieved competitive performance, the error of the translation E_T ranked the second

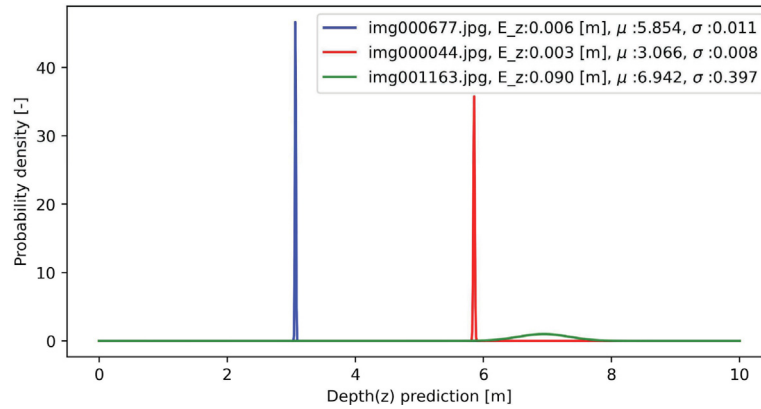
best, only slightly higher than that of STGC. Nevertheless, based on the above ablation study and comparison, the proposed PVSPE has shown outstanding generalization and robustness.

6.5. Uncertainty quantification of pose predict results

In addition to predicting the accurate relative pose of the spacecraft, PVSPE is also capable of estimating the uncertainty associated with the pose prediction. The importance of uncertainty estimation is twofold: it represents the confidence of prediction, facilitating fault analy-



(a) Pose prediction results, class activation maps and corresponding uncertainties of rotation



(b) Depth (z in 3D translation) prediction and corresponding uncertainty

Fig. 10. Qualitative results on sunlamp test domains of SPEED + dataset. (a) The first column visualizes the pose prediction results: the red and blue axes (arrows) represent the prediction and the true values, respectively. The second column visualizes class activation maps of the network. The third column visualizes the uncertainty of rotation by the method in (Yin et al., 2023): the red axes indicate the prediction of the rotation matrix, which is derived from the maximum likelihood estimate of the predicted F matrix, and the black axes represent the ground truth of 3D rotation. The more concentrated the red color region of the sphere, the smaller the rotation uncertainty. (b) The horizontal axis represents the depth (z in 3D translation) prediction value and the vertical axis represents the probability density value of depth prediction distribution (1D Gaussian distribution). The steeper the curve, the smaller the depth uncertainty.

sis; second, it is also a key input for the navigation system. Therefore, to quantify the uncertainty of the pose distribution, we use entropy information to indicate the degree of disorder or randomness of a closed system. Given $f_R \sim \mathcal{MF}(\mathbf{R}; \mathbf{F})$, the entropy H_R of 3D rotation is as follows:

$$H_R = H_{\mathcal{MF}(\mathbf{R}; \mathbf{F})} = - \int_{R \in SO(3)} f(R) \log f(R) dR \quad (30)$$

About the detailed derivation of $H_{\mathcal{MF}(\mathbf{R}; \mathbf{F})}$, for the sake of conciseness and clarity, please refer to (Yin et al., 2023).

Table 8

Run-time performance of PVSPE on Nvidia Tesla T4 (T4) and Nvidia Jetson Xavier NX (NX).

Platform	Time [s]	lightbox			sunlamp		
		$E_T[-]$	$E_R[\text{rad}]$	$E_{\text{pose}}[-]$	$E_T[-]$	$E_R[\text{rad}]$	$E_{\text{pose}}[-]$
T4	0.032	0.017	0.084	0.101	0.022	0.156	0.178
NX	0.384	0.017	0.084	0.101	0.022	0.156	0.178

Table 9

The initial mean orbital elements of the servicer.

OE	$a[\text{km}]$	$e[-]$	$i[^\circ]$	$\Omega[^\circ]$	$\omega[^\circ]$	$M[^\circ]$
Servicer	7078.135	0.001	98.2	189.9	0	0

The 3D translation entropy H_T can be represented in the same way. Given $f_T \sim \mathcal{M}\mathcal{G}(\mathbf{T}; \Sigma)$, H_T can be followed as:

$$H_T = H_{\mathcal{M}\mathcal{G}(\mathbf{T}; \Sigma)} = - \int_{T \in \mathbb{R}^{1 \times 3}} f(T) \log f(T) dT \quad (31)$$

$$H_T = \frac{3}{2} + \frac{3}{2} \log(2\pi) + \frac{1}{2} \log(|\Sigma|) \quad (32)$$

As shown in Fig. 9 (a) and (b), we drew the relationship between pose error and the corresponding distribution entropy. It shows that distribution entropy effectively captures the pose prediction error. A lower entropy generally indicates a more peaked distribution which exhibits higher confidence, less uncertainty and error. By setting a proper entropy tolerance threshold, the system can accept major-

ity inlier prediction, as well as reject outlier predictions with high-value error and perform re-actions.

6.6. Visualization analysis of the prediction results

Depicted in Fig. 10 (a) and (b), we further visualized the pose prediction result, class activation map (CAM), rotation uncertainty and depth (z in 3D translation) uncertainty for three typical images with labels 000677, 000044 and 001163 on the challenging sunlamp test domain. The CAM (Zhou et al., 2016) can visualize the attention regions that contribute to spacecraft pose estimation. As shown in the second column of Fig. 10 (a), the attention regions of the network under different illuminations and MLI reflections on the sample images can be observed.

In the first two rows of Fig. 10 (a), on img000677 and img000044, benefiting from the two self-attention mechanisms (SRA and ECA), the proposed PVSPE focuses on antenna, panel and some corner regions, thereby accurately predicting the spacecraft pose under strong noises. This is

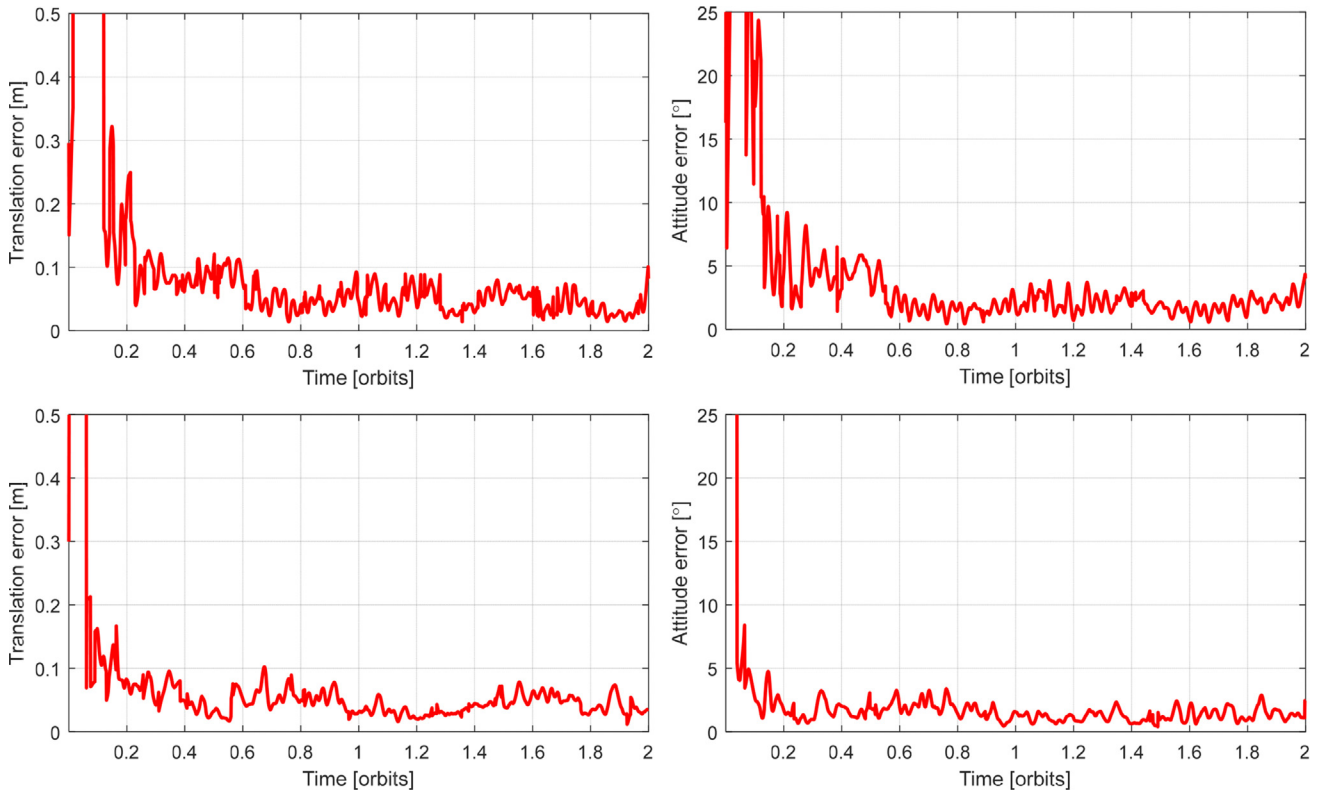


Fig. 11. Translation and attitude errors of vision navigation pipeline on SHIRT dataset. Scenario 1(Top). Scenario 2(Bottom).

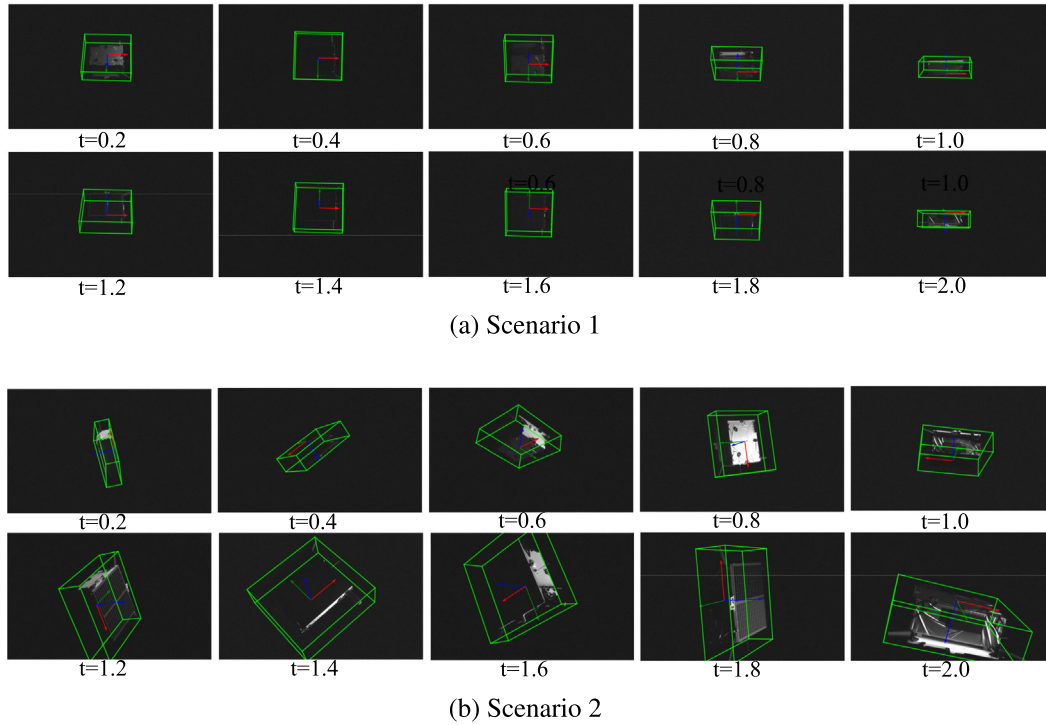


Fig. 12. Visualization of pose estimation results in two typical rendezvous scenarios on SHIRT dataset. The unit of time (t) is the orbit period. (a) Scenario 1 (b) Scenario 2.

logical since the relative position and orientation of the spacecraft can be determined by these components. In some extreme cases, with high-intensity local reflection on MLI, like the last row of Fig. 10 (a), the number of effective attention regions drops greatly, leading to larger pose errors and uncertainties.

6.7. Runtime performance

To further evaluate the proposed algorithm, we deployed it both on Nvidia Tesla T4 (T4) and Nvidia Jetson Xavier NX (NX). Tesla T4 has been employed in Spaceborne Computer-2 (SBC-2) (Swope et al., 2023) on the International Space Station (ISS). NX is a credit-card size and low-power (10 W/15 W) system-on-a-chip (SoC). Such a small size and power makes the NX capable of being deployed on a satellite, or as the primary instrument of a CubeSat.

The resolution of the input image is 768×512 . The model is exported in TensorRT¹ format. As shown in Table 8, the average inference speed of PVSPE on platforms T4 and NX is about 0.032s and 0.384s per image respectively. The result demonstrated that PVSPE has a near real-time performance.

6.8. Performance of PVSPE-based vision navigation pipeline

In addition to assessing the performance of PVSPE, we also evaluate the effectiveness of the PVSPE-based navigation pipeline on the challenging SHIRT dataset. The SHIRT dataset (Park and D'Amico, 2023) contains two typical rendezvous scenarios in low-earth orbit (LEO). Scenario 1 introduces an along-track separation trajectory following the standard v-bar hold point. Scenario 2 depicted a slow approach motion along a small relative semi-major axis trajectory. The state propagation occurs every second and lasts for two full orbital periods. The initial mean orbit elements (OE) of the servicer are listed in Table 9. The angular velocity of the target about its principal axes is initialized to $[1, 0, 0]^T (^\circ/s)$ for scenario 1 and $[0, 0.4, 0.6]^T (^\circ/s)$ for scenario 2. As shown in Fig. 11 and Fig. 12, based on pose prediction and uncertainty from the PVSPE network, the filter converges to low error even at a large initial disturbance. In the steady state, the attitude error is kept below 4° , whereas the translation error is to the centimeter level, indicating that the initial pose measurements provided by PVSPE are both accurate and reliable. Moreover, the pose distribution fitted by the PVSPE network effectively reflects uncertainty resulting from severe orbital illumination conditions.

7. Conclusion and future works

In this paper, we proposed a novel ViT-based multitask probabilistic network for monocular spacecraft pose esti-

¹ <https://developer.nvidia.cn/tensorrt>

mation. An enhanced PVT module is designed to combine global spatial and channel self-attention. A simple yet more efficient GFPN module is proposed for feature fusion. In order to precisely regress pose and model uncertainty, Matrix Fisher and Multivariate Gaussian distributions were employed. We further integrated PVSPE with Kalman filters to establish a vision-based navigation pipeline. We evaluated the proposed method on the challenging SPEED + and SHIRT datasets, and the superior results demonstrate that the proposed PVSPE and vision navigation pipeline accurately predict the spacecraft pose in various challenging scenarios. Like other DL-based SPE methods, it has some limitations in real-time performance on some resource-limited spacecraft hardware. In the future, a lightweight version of the current network architecture will be investigated to increase inference speed, such as a hardware-friendly vision attention mechanism.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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